

Clinic Appointment Management System

CSCI 101 – Final Project (Phase 4)

1. Problem Description

Managing patients and appointments manually in small clinics is time-consuming and prone to errors. Doctors and receptionists may struggle with:

- Tracking patient information efficiently
- Booking appointments without conflicts
- Displaying schedules quickly

The **Clinic Appointment Management System**, developed in Python, automates these processes. It allows clinics to organize patient records and appointments using **file handling** and **modular programming**, offering both a **command-line interface** and a **GUI** for ease of use.

2. Work Distribution

Team Member	Responsibilities
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Amira Mohamed	Patient management functions, file handling (patients.txt), conflict detection, and update operations
Mariam Mahmoud	Appointment booking, cancel operations, and GUI design using CustomTkinter

3. How the Program Solves the Problem

The system uses external text files (patients.txt and appointments.txt) to store patient and appointment data, making sure the data is saved and not lost. It is organized into **modular functions**, each responsible for a specific task:

- Adding and saving patient information
- Booking and updating appointments

- Canceling appointments
- Displaying full clinic schedules

Conflict detection ensures no two appointments are booked on the same date and time. The **GUI** provides an interface for clinic staff, while the **command-line menu** supports basic operations and testing.

4. How to Use the Program (User Manual)

1. Run the program using Python.
2. Choose an interface:
 - Command-line menu
 - GUI
3. Available features:
 - Add and save patients
 - Book appointments
 - Update appointment details
 - Cancel appointments
 - Display full clinic schedule